Unit 8 | Chapter 13: The "Network" (APIs, JSON & Fetch)

Learning Targets:

- Technical: I can explain the Request/Response cycle and identify the roles of the Client (Browser) and the Server. (NV 7.2.1)
- Technical: I can demonstrate the use of data exchange formats by parsing and stringifying JSON data. (NV 7.1.2)
- Technical: I can utilize the Fetch API and Async/Await syntax to retrieve data from external REST APIs. (NV 7.2.1)
- Technical: I can implement robust error handling using try/catch blocks and check for HTTP response status. (NV 7.2.1)
- Employability: I can demonstrate analytical reasoning and cognitive flexibility to interpret third-party API documentation and troubleshoot network-related bugs. (WRS 1.2.3)
- Employability: I can demonstrate probity and professional decorum by securely handling API keys and respecting the Rate Limits and Terms of Service of external providers. (WRS 1.1.2, 1.1.4)

Compelling Question:
How do modern applications connect to a global network of data to provide real-time information to users?

Score	Performance Indicators
4.0	**I can:** ☐ Architect a multi-step asynchronous workflow where data from one API is used to trigger a request to a second API, demonstrating advanced analytical reasoning. (NV 7.2.1, WRS 1.2.3) ☐ Implement an advanced loading and error UI state that provides specific feedback to the user based on HTTP error codes (e.g., 404 vs 500), displaying profound cultural competency and empathy. (NV 7.2.1, WRS 1.1.5) ☐ Maintain strict probity by utilizing environment variables to protect sensitive API keys from being pushed to public repositories. (WRS 1.1.2)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	**I can:** ☐ Successfully use fetch() to retrieve a JSON object and render its properties into the DOM using systematic resolution. (NV 7.2.1, WRS 1.2.3) ☐ Use async and await keywords correctly to handle the asynchronous nature

	of network requests. (NV 7.2.1) ☐ Parse complex nested JSON objects into usable JavaScript variables. (NV 7.1.2) ☐ Implement a try/catch block to prevent the application from crashing during a network failure, exhibiting professional decorum. (NV 7.2.1, WRS 1.1.4) ☐ Demonstrate intrinsic motivation and punctuality by completing daily networking labs on schedule. (WRS 1.1.1)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	**I can** ☐ Define common terminology and identify syntax: ☐ API, JSON, Request/Response, URL Parameters, Endpoint, fetch, .then(), async, await, try/catch, HTTP Status Codes (200, 404, 500). (NV 7.2.1, 7.1.2) ☐ Identify the importance of punctuality (WRS 1.1.1), probity in handling API access (WRS 1.1.2), professional decorum in technical naming (WRS 1.1.4), inclusive collaboration (WRS 1.1.5), and applying analytical reasoning to network logs (WRS 1.2.3).
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.